

Customer Churn Prediction for Telecom

AI & ML Internship Project Report

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1. Introduction

Customer churn refers to customers discontinuing a telecom service. Predicting churn helps companies retain valuable customers through targeted retention strategies. This project builds machine learning models to identify customers likely to churn.

2. Objectives

Load and clean the dataset; perform exploratory data analysis (EDA); encode categorical variables; train Decision Tree and Gradient Boosting models; optimize performance using cross-validation and hyperparameter tuning; identify important features.

3. Dataset

The Telco Customer Churn dataset from Kaggle contains customer demographics, subscription details, billing information, tenure, monthly charges, total charges, and the target variable 'Churn'.

4. Data Preprocessing

Removed customerID, converted TotalCharges to numeric, filled missing values, converted Churn to numeric, and applied one-hot encoding.

5. Exploratory Data Analysis

EDA included churn distribution, monthly charges distribution, contract type vs churn, tenure distribution, and a correlation heatmap. Insert your graph screenshots in this section before final submission.

6. Machine Learning Models

Decision Tree and Gradient Boosting classifiers were trained. Cross-validation and GridSearchCV were used for model optimization.

7. Model Evaluation

Evaluate the models using Accuracy, Classification Report, Confusion Matrix, Cross-validation Score, and Best Hyperparameters. Replace this placeholder with your notebook results.

8. Feature Importance

Use the optimized Gradient Boosting model to identify the most influential features affecting customer churn. Insert the feature importance chart.

9. Conclusion

The project demonstrates the complete machine learning workflow from preprocessing to model evaluation. Gradient Boosting generally achieved better predictive performance than the Decision Tree model.

10. Future Scope

Future work could include Random Forest, XGBoost, handling class imbalance, and deploying the model as a web application.